

BioForms: Exploring the Materiality and Interactive Affordances of Algae-based Bioplastics

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Figure 1: Versatile affordances of algae-based bioplastics: agar screenprinting, laser cut reactive patterning, spongy foam structure, and 2.5D poured membranes.

Abstract

For a long time, the primary focus of bioplastics has been to replace petroleum-derived plastics. As a result, current fabrication processes rarely adapt to the inherent characteristics of bioplastics, failing to showcase their true capabilities. This paper offers a series of unique properties of algae-based bioplastics uncovered through the Material-Driven Design framework [12], and fabrication processes that reveal novel affordances. We leverage properties often considered limitations such as shrinkage, heat and moisture reactivity, swelling, and long drying times to develop four fabrication techniques: agar screenprinting, laser cut reactive patterning, spongy foam structure, and 2.5D poured membranes. By utilizing commercially available ingredients and accessible tools, we aim to lower the barrier to exploring biomaterials in DIY settings. We advocate for understanding a material's inherent characteristics as a first step in research and development, moving the focus beyond a human-centered to a material-led approach, further inspiring novel and better-suited applications.

*Both authors contributed equally to this research.

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CCS Concepts

• Human-centered computing → Systems and tools for interaction design.

Keywords

Digital Fabrication; Biomaterials, Bioplastics, Shape-changing Materials, Tangible Interfaces and Affordances, Human-Plant Interaction, DIY Practices, Sustainability, Material-Driven Design

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1 Introduction

The environmental crises caused by non-degradable materials have raised interest in researching materials with lower environmental impacts. Bioplastics, specifically, have been gaining attention due to their degradability and tunability, which is perceived to offer a promising solution to substitute petroleum-based plastics. However, unlike the development of other materials driven by their capabilities [21], the inherent strength of bioplastics is often underexplored due to this perception. Bioplastics are processed with unsuitable manufacturing techniques often used for traditional plastics, with the expectation that they need to perform identically to non-degradable plastics [1, 7]. This misalignment results in the underperformance and misconception of bioplastics, preventing

research from progressing beyond the constant doubts regarding their ability to replace non-renewable plastics.

In the HCI field, studies around biomaterials have shifted focus to Do-It-Yourself (DIY) and fabrication aspects. Alganyl [2] introduces an easy DIY technique to produce bioplastics at home and an overview of innovative affordances. Koelle et al. [13] leverage the customizability and compatibility of bioplastics to different materials (e.g. carbon or charcoal powders) to create conductive devices, expanding the applications of bioplastics in the interaction design field. However, those projects usually position bioplastics as a medium to showcase the properties of other materials, not centered around the strength of bioplastics itself.

Motivated by the Material-Driven design method [12] and prior works, this research aims to explore the undiscovered properties and affordances of bioplastics. Rather than matching the performance of bioplastics to petroleum-based plastics, we take a step back in unfolding and understanding the inherent properties of bioplastics as an independent material. We selected an algae-based bioplastic using commercially-ready materials and tools, as it lowers the barriers and enables a deeper hands-on engagement. Inspired by its unique properties such as shrinkage, heat and moisture reactivity, swelling, and long drying times, we developed fabrication processes tailored to those properties and demonstrated new affordances that transform limitations into strengths. We present BioForms, a series of novel affordances and the corresponding fabrication methods highlighting algae-based bioplastics' distinct material properties. Rather than being application-focused, this paper provides an overview of the fabrication and interaction possibilities of algae-based bioplastics. By showcasing the versatility and scalability of our approach, we aim to inspire other practitioners to build on our work, moving beyond human-centered to a material-led perspective on creating well-suited applications.

2 Related Work

2.1 Bio-HCI and Crafting with Biomaterials

Recently, biomaterials have become a popular field in HCI research due to sustainability concerns. Compared to the broad definition of biomaterials that includes naturally grown materials such as wood, plant fibers, and metals, there is an increased interest in crafting biomaterials from naturally derived polymers such as gelatin [14], chitosan [9], starches [4], etc. Unlike the *grown biomaterials* that have predefined material properties, these *crafted biomaterials* require secondary processing of raw ingredients, enabling individuals to tinker with and customize ideal material properties. Researchers utilized this advantage in creating interactive artifacts: Organic Primitives[11] demonstrates a series of environmental actuations using chitosan and organic compounds. Koelle et al. [13] further utilize the compatibility of *crafted biomaterials* to create soft, interactive I/O devices. Bell et al. [5] have explored bio-making in DIY settings. These practices showcase a wide variety of potentials in *crafted biomaterials*. However, these interactions usually highly rely on the additives, such as photochromic powders [2] or stainless steel fibers [14], not focusing on the properties of biomaterials themselves. In addition, only limited fabrication techniques are employed in these practices, such as sheet-casting, molding, or

hand-shaping, restricting the artifacts from showing diverse form factors and behaviors.

2.2 Material-Driven Design

In contrast to an application-oriented approach, we adopted the Material-Driven Design (MDD) framework [12], which proposes understanding the “material experience” first to find out the best suitable applications. For example, Loop Biotech [6] utilizes the ability of mycelium to decompose dead organic matter to design coffins that can accelerate the degradation. In addition, the MDD framework emphasizes the hands-on, crafting [16] engagement with the material, letting the sensory experience naturally inform the design development. For instance, Bio-Digital Calendar [3] makes use of the physical changes during the growing process of kombucha SCOBY to create sound experiences.

In our practice, rather than focusing on certain applications for bioplastics (e.g. replacing plastic packaging) or emotional experience, we started by experimenting with and understanding the diverse material characteristics (physical, mechanical, and chemical properties) of bioplastics and developed design attempts based on the findings.

2.3 DIY and Accessible Making

Influenced by the rising Maker culture [20], democratizing design has become the goal of researchers in the HCI community. Online biomaterial libraries such as Materiom [15] and Material Factors [10] enable individuals to replicate and create their own bioplastics using off-the-shelf ingredients at home. These resources give the opportunity for non-scientists to understand and tinker with *crafted biomaterials*, shifting material innovation beyond the confines of scientific laboratories. However, these tutorials often end up making 2D “sheet-like”, non-functional samples with inconsistent quality, and a lack of variations in fabrication processes. Building on this, our practice aims to create diverse processing techniques using accessible ingredients and tools, lowering the barrier to bio-making while expanding knowledge in fabricating biomaterials.

3 Exploration and Development

3.1 Biopolymer Selection and Processing

When choosing the raw materials for the *crafted biomaterials* to be explored, we considered a variety of options, including biopolymers sourced from vegetable starches, seaweed/algae, gelatin, and fermented sugars. We selected red algae (or agar agar) as our primary biopolymer source for investigation, which is a species native to the Pacific Coast. Agar is a biopolymer derived from the cell walls of red algae, consisting of two polysaccharides: agarose and agarose pectin [18]. It dissolves in hot water. When heated above its melting point of 90°C and subsequently cooled, forms a gel matrix. This matrix results from the crosslinking of agarose polymer chains through hydrogen bonding [19].

We landed on exploring algae-based bioplastics due to their:

Simplicity: It requires only three ingredients: agar (from red algae) as the polymer, vegetable glycerin as the plasticizer, and water as the solvent.

Accessibility: All materials are readily available in grocery stores, and the material is well developed, with extensive online resources providing guidance on using agar to create bioplastics.

Adjustability: The material's properties can be easily fine-tuned during production. For example, adjusting the glycerin ratio alters its plasticity, while varying the pour thickness modifies its toughness and shrinkage characteristics.

In order to create the agar bioplastic, the ingredients are mixed together and heated to create an *agar solution*. Once the solution is heated above 90°C, it begins to polymerize and forms an *agar gel*. It is poured into a mould, and as the water evaporates, the gel dries into an *agar bioplastic* (See Appendix fig.6 and fig.7). During our explorations, we observed several key characteristics and intrinsic properties of the *agar bioplastic*:

Gel State: Agar exhibits unusually high hysteresis properties, forming a gel when cooled below 34° and maintaining this semi-solid state unless reheated above 85–90°C. Its ability to gel within a few minutes and remain stable at room temperature provides a unique opportunity for design applications in the semi-solid state [22].

Shrinkage: Known for its ability to retain large amounts of water, agar exhibits a phenomenon referred to as "agar swell". Therefore, agar tends to shrink during drying as water evaporates from its internal structure [17].

Degradability and Recyclability: Agar is biodegradable through specific soil bacteria and composting methods, including thermal or enzymatic decomposition. Additionally, it can be remelted above 90°C and recast, enabling recyclability and circularity [8].

Insolubility in Cold Water: Agar is insoluble in cold water and, once gelled, does not melt unless heated above 85–90°C. This ensures its stability in aquatic environments without degrading. However, its hydrophilic nature allows it to swell significantly in cold water, absorbing over 20 times its weight in water [18].

Heat and Moisture Sensitivity: Agar bioplastic films are sensitive to heat and moisture. Heating the dried film can cause creasing and shrinkage, while exposure to high-humidity environments causes it to swell and expand.

3.2 Exploration of Fabrication Methods and Affordances

Our experimentation was primarily guided by the MDD methodology—observing the materials' natural tendencies and behaviors to develop fabrication methods. In addition, we brainstormed and referenced existing parallel methods to test suitable techniques. For instance, the natural shrinkage and curling of the agar bioplastic under sunlight inspired us to explore its heat-reactive properties further. We also found that the gel state offered a controlled way to manipulate shapes while still achieving uniform results. This discovery led us to innovate with techniques like cutting, stamping, and stacking the material, along with experimenting with screen printing. The high viscosity of the agar gel allowed it to adhere to 3D surfaces, enabling us to explore techniques like spraying and pouring it onto complex structures.

By building on these unexpected observations, we achieved intriguing results through hands-on experimentation, examining how the material could be shaped, manipulated, and potentially applied across various design contexts.

3.3 Agar Screenprinting

Traditionally, screen printing typically uses inks and paints, however we sought to explore the potential of screen printing with agar gel inspired its high viscosity. We experimented with various tools, including both traditional mesh screens and alternative options such as vinyl and silicone stencils, on fabrics like cotton, polyester, and nylon.

We found that when the agar began forming a gel structure, it had the optimal consistency for screen printing. This gel state allowed the agar to hold its shape and prevent leakage, while still maintaining a spreadable viscosity. We also discovered that the most effective method was directly screen printing with a 1/25" silicone stencil on cotton or other natural fabrics.

Minimum Feature Size: To determine the smallest printable features, we used a laser cutter to create a gradient of sizes (see table below). We found that features smaller than this led to inconsistent, faded, or peeling prints.

Heat Reactivity: As the algae-based prints dried on fabric, they shrank, creating a textured, cinched effect, particularly noticeable on the unprinted side. This shrinkage can add unique textures or functional cinching to specific fabric areas. Applying heat above 70°C with a heat gun further enhanced the tactile and visual patterns by increasing the shrinkage.

Temporal Features: Thanks to agar's low melting point, the prints can be removed by boiling the fabric or washing it in hot water. This makes agar prints ideal for temporary designs that perform well in dry conditions but are easily washable and removable when desired.



Figure 2: (a) Process of screenprinting agar using a silicone stencil. (b) Agar screenprinting in gel state before drying. (c) The cinching effect due to agar shrinkage after drying. (d) Original screenprints (dyed in blue). (e) The screenprints dissolved after washing in boiling water. (f) Minimum feature sizes for various geometries.

3.4 Laser Cut Reactive Patterning

Designing cut patterns on a 2D bioplastic surface allows us to harness environmental changes such as heat and moisture to generate 2.5D reactions and movements, resulting in diverse features and textures within the material. The agar bioplastic was patterned by laser cutting in order to create visual effects when the material shrinks. Then we actuated the reactions using a heat gun or a humidifier. The shrinkage behavior can be designed through different patterns (see table below). These effects can be achieved at scale through digital fabrication, unlocking new opportunities for dynamic, responsive, and interactive sensing applications with agar bioplastics.

Irreversible Heated Shrinkage: When exposed to 80°C heat using a heat gun for 4–8 seconds, the material undergoes a one-time, irreversible shrinkage and creasing along the cut lines. This process increases the stiffness of the heated material and creates

distinctive "curling" or "spiky" effects. By adjusting cutting patterns, this technique enables control over surface porosity and adds rigidity, offering enhanced texture for both aesthetic and functional applications.

Humidity-Sensitive Actuation: Placing the laser cut material approximately 2 inches away from a humidifier creates a localized high-humidity environment, triggering reversible actuation along the cut lines. Due to the slight absorption of water along one side of the material, it expands, creating this actuation, and when removed from the high-humidity environment, the water dries out, returning the material to its original state. This provides a low-energy approach to enable environmental sensing through material movement in architectural or exhibition settings.

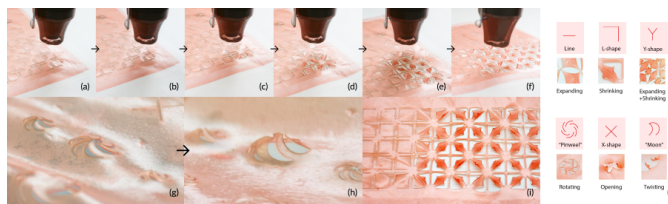


Figure 3: Figure 4: (a)-(f) The heat actuation process. (g) Before applying moisture. (h) After applying moisture for 2 mins. (i) A transition from the original cut-outs (left) to heated patterns (right). (j) Patterns and the corresponding reactions to heat.

3.5 Spongy Foam Structure

In order to create a three-dimensional volume in the agar bioplastic material, we evenly distributed degradable soap bubbles into the agar solution to develop a lightweight agar foam (See Appendix fig.8 and fig.9). It is an ultra-lightweight 3D foam structure with flexibility and irreversible compression properties, enabling it to be moulded and shaped easily to create various forms.

Buoyancy: The foam structures are naturally buoyant, floating on the water's surface even after swelling and short-term exposure to cold water. This buoyancy enables them to retain their form while submerged for 5–7 days, making them ideal for applications requiring temporary durability in aquatic environments before safely degrading.

Irreversible Compressibility: The spongy cross-section of the agar foam enables it to compress irreversibly under high pressure. Laser cut acrylic stamps were used to create this embossing effect. This property allows the foam's surface to be easily patterned, creating forms that can be molded to objects while adding distinct tactile and visual textures.



Figure 4: (a) A piece of floating agar biofoam (left) and a block of agar gel sinking to the bottom (right). (b) Side view of image (a). (c) Brain coral inspired pattern imprinted on the agar biofoam. (d) The 2.5D effect of compressing agar biofoam.

3.6 2.5D Poured Membranes

Agar films can be formed on 3D surfaces by pouring the agar solution over textured forms, capturing surface details with high resolution. While spraying the agar solution onto 3D shapes like a bowl adhered well to uneven surfaces, the rapid gelling properties often clogged the spray mechanism. To explore more complex geometries, we used Grasshopper to design parametric molds for 3D printing and created Origami patterns that were cut and creased on the Zund. Pouring the agar solution onto these non-flat surfaces allowed it to dry into desired 2.5D shapes.

3D Shape Retention: When cast in a 3D mold, the agar-based film dries while maintaining the mold's shape, textures, and intricate details. This property facilitates the creation of pre-formed structures with minimal post-processing, ideal for degradable membranes that conform to organic or complex forms.

Tunable Transparency: As the agar solution dries on uneven surfaces, gravity redistributes the material, resulting in varying thicknesses that follow the mold's contours. These thickness variations create optical effects that can be further enhanced with added coloring agents. Using parametric molds, we can achieve diverse light-filtering patterns and aesthetic effects in the bioplastic.



Figure 5: (a) The casting process on an Origami piece. (b) Agar bioplastic sheet dried in the Origami form. (c) Various parametric patterns demonstrating the color distribution effect. (d) The 3D printed molds used for casting the agar bioplastic sheet.

3.7 Scanning Electron Microscopy Analysis

The microstructures of the agar bioplastic were analyzed to gain a deeper understanding of the changes occurring at the micron scale. The TM4000Plus SEM was used to examine both the top surface and cross-sections of the samples.

Agar Bioplastic: The agar bioplastic sheet features a smooth top surface, as shown in Fig 6 (a).

Heated Agar Bioplastic: Heating the agar bioplastic significantly alters its surface structure, explaining its shrinking behavior. Creases on a microscale (approximately 10–100 μm between creases) indicate the material's directional shrinkage, influenced by the laser cut lines.

Agar Foam: The agar foam shows a smoother top surface with small scattered pores no larger than 0.8 mm in diameter. Its cross-section has a spongy texture, with pores ranging from approximately 0.5–8 mm in size. The spongy walls formed by agar bioplastics are extremely thin, measuring in the tens of microns, which contributes to the material's lightweight and buoyant characteristics.

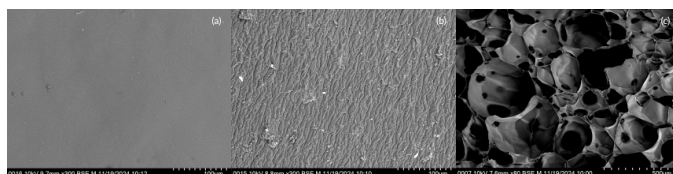


Figure 6: Scanning electron microscopy images of (a) top surface of agar bioplastic sheet; (b) top surface after heated; (c) cross-section of agar biofoam.

4 Discussion

4.1 Transforming Constraints into Features—Manifestation of MDD

During the exploration phase, we noticed several constraints that hindered us from realizing design intentions. For example, we were planning to spray-paint the agar solution but kept encountering clogging due to its quick gelling property. However, following the MDD method, we decided to embrace the material's natural properties. We discovered that the gel state allowed shape manipulations in a more controlled way, creating uniform results. This drove us to improvise on cutting, stamping, and stacking the material, in addition to our trials of screen printing. The higher viscosity of the agar gel also enables it to adhere to 3D structures such as Origami. Our practice demonstrated how MDD can turn the perceived disadvantages into unique characteristics, unfolding the potential of emerging materials. We encourage designers to use this observation-driven and reflective approach in their research of material exploration.

4.2 Introducing New Perceptions of Bioplastics

Through this project, we hope to challenge the public perception of crafted biomaterials being uninviting, scrappy, and even “compost-like”. We addressed this issue by using parametric design and digital fabrication to increase the uniformity of designs. Natural food dyes also were added to make the outcomes more visually appealing. In addition, we showcased different surface features, including colors, opacity, and textures, which enables more artistic expressions in bio-design. Furthermore, to support the replication process, this paper contributes a detailed documentation of our techniques and

tools, aiming to increase the success of the results for other practitioners. We hope to reduce the obstacle and enhance people's trust in making crafted biomaterials.

4.3 Limitations

Although we embrace the distinct properties of agar bioplastics, they still bring some challenges regarding consistency and scalability. Since the agar bioplastic is highly sensitive to environmental conditions (e.g. room temperature, moisture, airflow, etc.), any fluctuations in these factors during the air-drying process can impact the consistency of outcomes. The shrinkage of agar bioplastics also constrains the maximum thickness of the dried bioplastic, as thicker pours will result in more shrinkage, limiting the dimensions of the material. This limitation might be overcome by adding fillers and expanding agents (such as soap bubbles in our practice). Lastly, since our practice was at a low-volume, DIY level, the scalability and manufacturability of our methods on a larger scale are still unknown. However, we believe that equipment at the laboratory/industrial level can provide a regulated environment for results with higher reliability.

4.4 Future Work

We plan to conduct user testing by facilitating workshops with designers and artists to evaluate the accessibility of our methods and observe whether the affordances can spark more unconventional creations with algae bioplastics. Crowdsourcing potential applications at public exhibitions could also effectively bring bio-based design closer to people's lives. On the technical side, specifications will be fine-tuned in order to develop functional and compelling applications. We also hope to further investigate the circularity aspect, as agar gel can be recycled and recast. This could provide promising solutions to the massive single-use waste or temporal installations, fostering a more comprehensive lifecycle. Lastly, by shedding light more on the material's inherent strength, we call for an emphasis on material-led design, focusing on what can be achieved by the material, not only for the benefit of humans. We envision this work can serve as a stimulus, inspiring other practitioners to create well-suited applications and extend innovations to other crafted biomaterials.

5 Conclusion

We introduce BioForms, four novel affordances and fabrication techniques centering on the inherent material properties of agar bioplastics. Inspired by the MDD method, we started with understanding the characteristics of agar bioplastics through hands-on exploration and experiments with external stimuli. Rather than competing with petroleum-based plastics, we embraced the natural materiality of bioplastics itself. Several unique properties of algae-based materials were identified during this process, such as the gel state, shrinkage, and heat/moisture sensitivity. They then were utilized to develop a series of affordances and the corresponding DIY fabrication methods, including agar screenprinting, laser cut reactive patterning, spongy foam structures, and 2.5D poured membranes. By contributing DIY methods using accessible ingredients and digitized tools, we aim to lower the barrier of crafting with bioplastics and challenge existing perceptions of biomaterials.

Lastly, instead of focusing on specific applications, our approach prioritized discovering the material's natural properties, technical functionalities, and developing suitable fabrication processes, allowing us to reveal unexpected possibilities in biomaterial design. We call for an emphasis on the material's inherent strength for future research on material innovations.

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A Recipes of agar bioplastics



Figure 7: Recipes used in this project.



Figure 8: Preparing process of agar bioplastics: (a) Step 1: Pour the measured ingredients into a pot. (b) Step 2: Stir continuously on medium heat until the viscosity increases, and volume reduces by 30%. (c) Step 3: Monitor the temperature to ensure the mixture reaches above 85°C. (d) Step 4: Add in natural dyes of your choice and mix thoroughly.

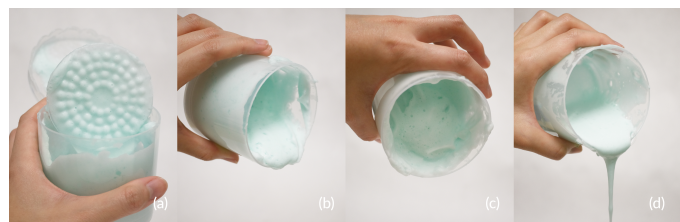


Figure 9: Example consistency of agar biofoam: (a) (b) Right Consistency: When the foam is “pourable” but has a viscous, whipped cream like texture. (c) Too Much Foam: Results in a foam that is too thick, and won’t pour. The molded result disintegrates easily. (d) Too Much Agar: Results in a foam that is too thin, and easily pours. The molded result is prone to significant shrinkage.

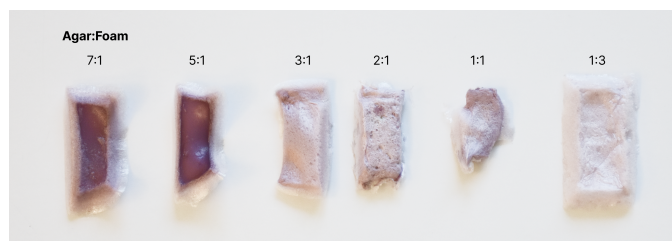


Figure 10: Volume ratio experiments of agar biofoam.

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